

```

1 // 7 & 1
2 // 111 & 001 = 001 => least significant bit is [1] so 111 is odd
3
4 // 6 & 1
5 // 110 & 001 = 000 => least significant bit is [0] so 110 is even
6
7 #include <iostream>
8 using namespace std;
9 int main() {
10     string name = "Ahmed Esam AbuHussien";
11     string section = "01-Network";
12
13     cout << name[0] << " " << name[6] << " " << name[11] << endl;
14
15     int num;
16     cout << "Enter an integer: ";
17     cin >> num;
18
19     cout << num;
20
21     if ((num & 1) == 0)
22     |     cout << " is Even\n";
23     else
24     |     cout << " is Odd\n";
25
26     return 0;|
27 }

```

Text Segment

Address	Instruction (Hex)	Source
0x00400000	0x3C011001	la \$t0, name
0x00400004	0x34280000	(part of la)
0x00400008	0x81040000	lb \$a0, 0(\$t0)
0x0040000C	0x2402000B	li \$v0, 11
0x00400010	0x0000000C	syscall
0x00400014	0x24040020	li \$a0, 32
0x00400018	0x2402000B	li \$v0, 11
0x0040001C	0x0000000C	syscall
0x00400020	0x3C011001	la \$t0, name
0x00400024	0x34280000	(part of la)
0x00400028	0x81040006	lb \$a0, 6(\$t0)
0x0040002C	0x2402000B	li \$v0, 11
0x00400030	0x0000000C	syscall
0x00400034	0x24040020	li \$a0, 32
0x00400038	0x2402000B	li \$v0, 11
0x0040003C	0x0000000C	syscall
0x00400040	0x3C011001	la \$t0, name
0x00400044	0x34280000	(part of la)
0x00400048	0x8104000B	lb \$a0, 11(\$t0)
0x0040004C	0x2402000B	li \$v0, 11
0x00400050	0x0000000C	syscall
0x00400054	0x24020004	li \$v0, 4
0x00400058	0x3C011001	la \$a0, newline
0x0040005C	0x34240047	(part of la)
0x00400060	0x0000000C	syscall
0x00400064	0x24020004	li \$v0, 4
0x00400068	0x3C011001	la \$a0, prompt
0x0040006C	0x34240021	(part of la)
0x00400070	0x0000000C	syscall
0x00400074	0x24020005	li \$v0, 5
0x00400078	0x0000000C	syscall
0x0040007C	0x00024021	move \$t0, \$v0
0x00400080	0x24020001	li \$v0, 1
0x00400084	0x00082021	move \$a0, \$t0
0x00400088	0x0000000C	syscall
0x0040008C	0x31090001	andi \$t1, \$t0, 1
0x00400090	0x11200005	beq \$t1, \$zero, even
0x00400094	0x24020004	li \$v0, 4
0x00400098	0x3C011001	la \$a0, odd_msg
0x0040009C	0x3424003E	(part of la)
0x004000A0	0x0000000C	syscall
0x004000A4	0x0810002E	j exit
0x004000A8	0x24020004	li \$v0, 4
0x004000AC	0x3C011001	la \$a0, even_msg
0x004000B0	0x34240034	(part of la)
0x004000B4	0x0000000C	syscall
0x004000B8	0x2402000A	li \$v0, 10
0x004000BC	0x0000000C	syscall

Data Segment

Address	Data
0x10010000	name
0x10010016	section
0x10010021	prompt
0x10010034	even_msg
0x1001003E	odd_msg
0x10010047	newline